

# Spatiotemporal Immune Landscape of Colorectal Cancer Liver Metastasis at Single-Cell Level



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**ABSTRACT**

Liver metastasis, the leading cause of colorectal cancer mortality, exhibits a highly heterogeneous and suppressive immune microenvironment. Here, we sequenced 97 matched samples by using single-cell RNA sequencing and spatial transcriptomics. Strikingly, the metastatic microenvironment underwent remarkable spatial reprogramming of immunosuppressive cells such as *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> M2-like macrophages. We further developed scMetabolism, a computational pipeline for quantifying single-cell metabolism, and observed that those macrophages harbored enhanced metabolic activity. Interestingly, neoadjuvant chemotherapy could block this status and restore the antitumor immune balance in responsive patients, whereas the nonresponsive patients deteriorated into a more suppressive one. Our work described the immune evolution of metastasis and uncovered the black box of how tumors respond to neoadjuvant chemotherapy.

**SIGNIFICANCE:** We present a single-cell and spatial atlas of colorectal liver metastasis and found the highly metabolically activated *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> M2-like macrophages in metastatic sites. Efficient neoadjuvant chemotherapy can slow down such metabolic activation, raising the possibility to target metabolism pathways in metastasis.

**INTRODUCTION**

Liver metastasis remains a major hurdle to long-lasting survival of patients with colorectal cancer (1, 2), which can be partly explained by the highly dynamic spreading routes of cancer cells (3, 4). Outside the cancer cells, the tumor microenvironment (TME) of liver metastasis harbors a highly immunosuppressive phenotype (5), induces a systemic loss of antigen-specific T lymphocytes (6), and drives the spread of tumors (7). It still remains largely unknown how the immune cells spatially orchestrate colorectal cancer liver metastasis (CRLM) progression and whether the metastatic cellular

microenvironment differs from the primary ones (8–13). There is hence a pressing need for recording, identifying, and quantifying cellular or even spatial landscape of CRLM to refresh our understanding of metastasis biology.

Neoadjuvant chemotherapy (NAC) refers to chemical drugs that are administered prior to surgical removal of a tumor. Paradoxical results have been seen for the effect of NAC in CRLM. In some reports, NAC was shown to prolong progression-free survival (14) and overall survival (15). However, especially in the patients with low-risk resectable CRLM, NAC was unexpectedly associated with an unimproved overall survival (16). Another independent group also reported that NAC did not prolong disease-free interval (17). Those seemingly contradictory data raise a critical but poorly understood question of whether NAC could favor antitumor response and improve outcome of patients with CRLM. Thus, dissecting the precise effect of NAC, especially TME alterations, is crucial for understanding the key mechanisms of NAC and designing novel therapeutic strategies.

Here, using single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics (ST) of 97 samples from 24 patients, we aim to explore the immune atlas of CRLM. We revealed that immune microenvironment showed dynamic cellular and spatial changes from primary to metastatic sites. Following NAC treatment, the immune phenotypes underwent antitumor remodeling in responsive patients but shifted toward more immunosuppression in nonresponsive patients, where the immunosuppressive cells were reprogrammed at metastatic sites. Our findings highlight the favorable effect of efficient NAC treatment in patients with resectable CRLM and allow for the data-driven design of novel therapeutic combinations such as NAC plus immunotherapy in selected patients.

**RESULTS****Integrated scRNA-seq and ST Precisely Quantify Immune Cell Diversity in Resectable CRLM**

To define the single-cell landscape of CRLM, we applied scRNA-seq and ST to quantify CD45<sup>+</sup> cell dynamics (Fig. 1A;

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**Note:** Supplementary data for this article are available at Cancer Discovery Online (<http://cancerdiscovery.aacrjournals.org/>).

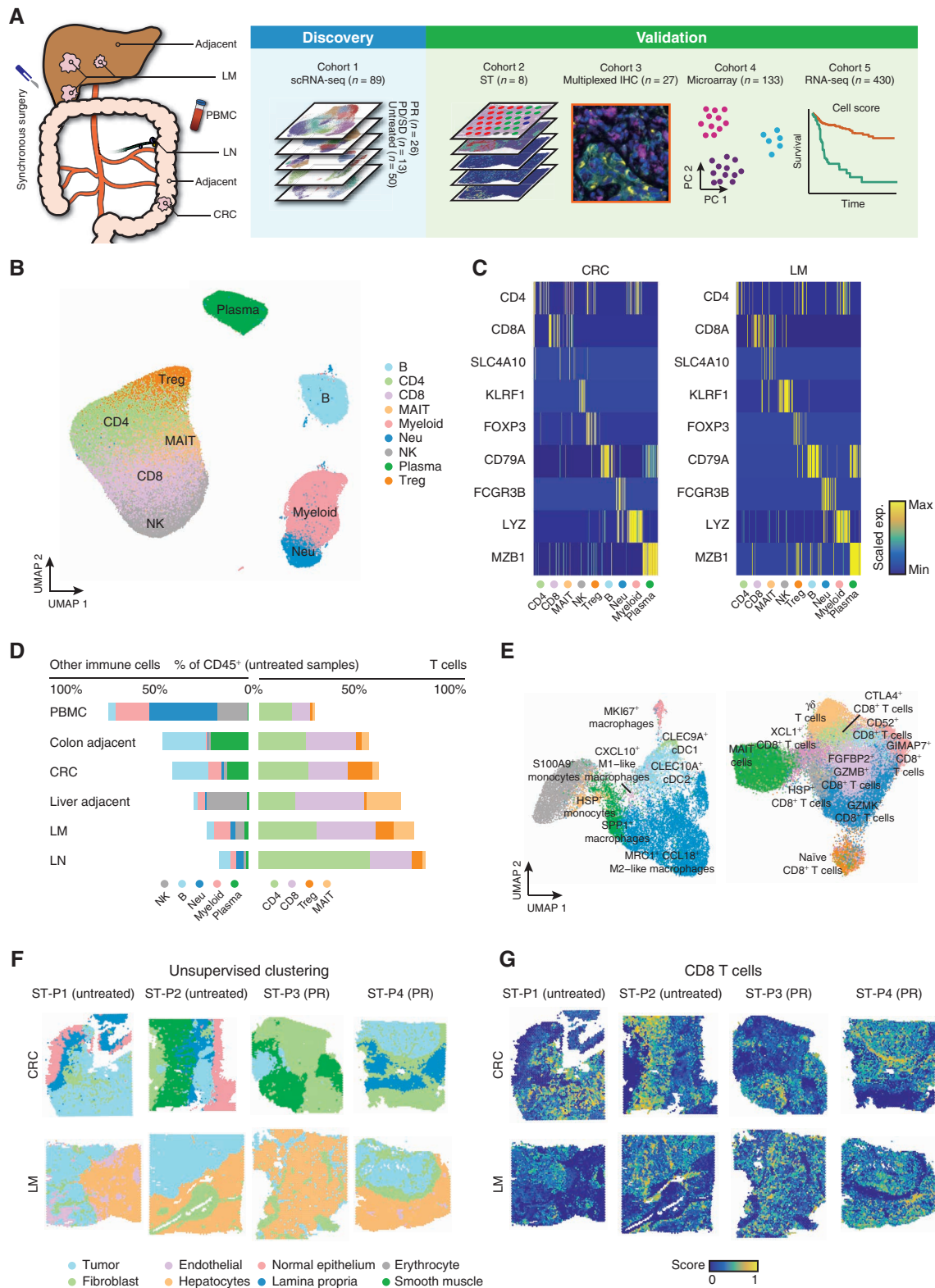
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**Figure 1.** scRNA-seq and ST revealed the immune landscape of CRLM. **A**, The experimental scheme for discovering and validating functional immune subpopulations in CRLM. CRC, colorectal cancer; LM, liver metastasis; LN, lymph node. **B**, The UMAP plot of all main immune cell types. CD4, CD4 T cell; CD8, CD8 T cell; MAIT, mucosal associated invariant T cell; myeloid, myeloid cell; Neu, neutrophil; Treg, regulatory T cell. **C**, The heat map of selected cell markers. Left, the immune cell in colorectal cancer; right, the immune cell gene-expression pattern in liver metastasis. **D**, Histogram showing the proportion of main immune cells across tissues. Right, the T-cell proportion; left, other immune cells. **E**, The UMAP plot showing the subpopulation of myeloid cells and CD8 T cells. **F**, The unsupervised clustering analysis of ST in colorectal cancer and LM. **G**, The CD8 T-cell score of ST in colorectal cancer and liver metastasis.

Methods) using paired samples of colorectal cancer, adjacent colon, liver metastasis, and adjacent liver, lymph nodes along colons, and peripheral blood mononuclear cells (PBMC). In total, 24 patients with resectable CRLM were recruited based on strict criteria (Supplementary Table S1). Resectability of the tumors has passed the multidisciplinary team (MDT) discussion composed by senior clinicians as described previously (18). In detail, 89 samples from 20 patients underwent scRNA-seq and 8 samples from 4 patients were sequenced by ST, among which 54 samples from 13 patients were treatment-naïve, 30 samples from 6 patients were treated with neoadjuvant chemotherapy with partial response (PR), and 13 samples from 5 patients were treated with neoadjuvant chemotherapy with progressive disease (PD) or stable disease (SD; Fig. 1A). All the tumors were microsatellite-stable.

After two rounds of quality control and doublet removal of scRNA-seq data (see Methods), 79,703 cells from untreated patients ( $n = 11$ ), 36,284 cells from NAC-PD/SD patients ( $n = 5$ ), and 62,643 cells from NAC-PR patients ( $n = 4$ ) were used for further analysis. On average, each sample contained 2,007 cells, and 1,064 genes (median) were quantified in each cell, with no obvious batch effects across different samples (Supplementary Fig. S1A). We integrated all 178,630 CD45<sup>+</sup> cells, performed the clustering analysis, and defined the main cell types using the SingleR (19) and cell marker genes (Supplementary Table S2; Methods). Largely consistent with previous scRNA-seq of colon cancer (8, 20), we identified myeloid cells, CD8<sup>+</sup> T cells, CD4<sup>+</sup> T cells, natural killer (NK) cells, and B cells from the CRLM samples (Fig. 1B and C; Supplementary Fig. S1A–S1C). Of note, T-cell proportion was obviously distinct across different tissue types, suggesting the tissue heterogeneity of the immune microenvironment (Fig. 1D). Both colorectal cancer and paired liver metastasis showed ~10% regulatory T cells (Treg) of all CD45<sup>+</sup> cells, whereas such cells were scarce in the adjacent normal tissues (Supplementary Fig. S1D and S1E). This indicated a suppressed intratumor immunity, which was previously ascribed to the reduced immunotherapeutic efficacy in liver metastasis (21). Likewise, we integrated the SingleR (19) and manual marker-based annotation to further define immune cell subsets (Fig. 1E; Supplementary Fig. S1F; Supplementary Table S2; see Methods) for further in-depth analysis.

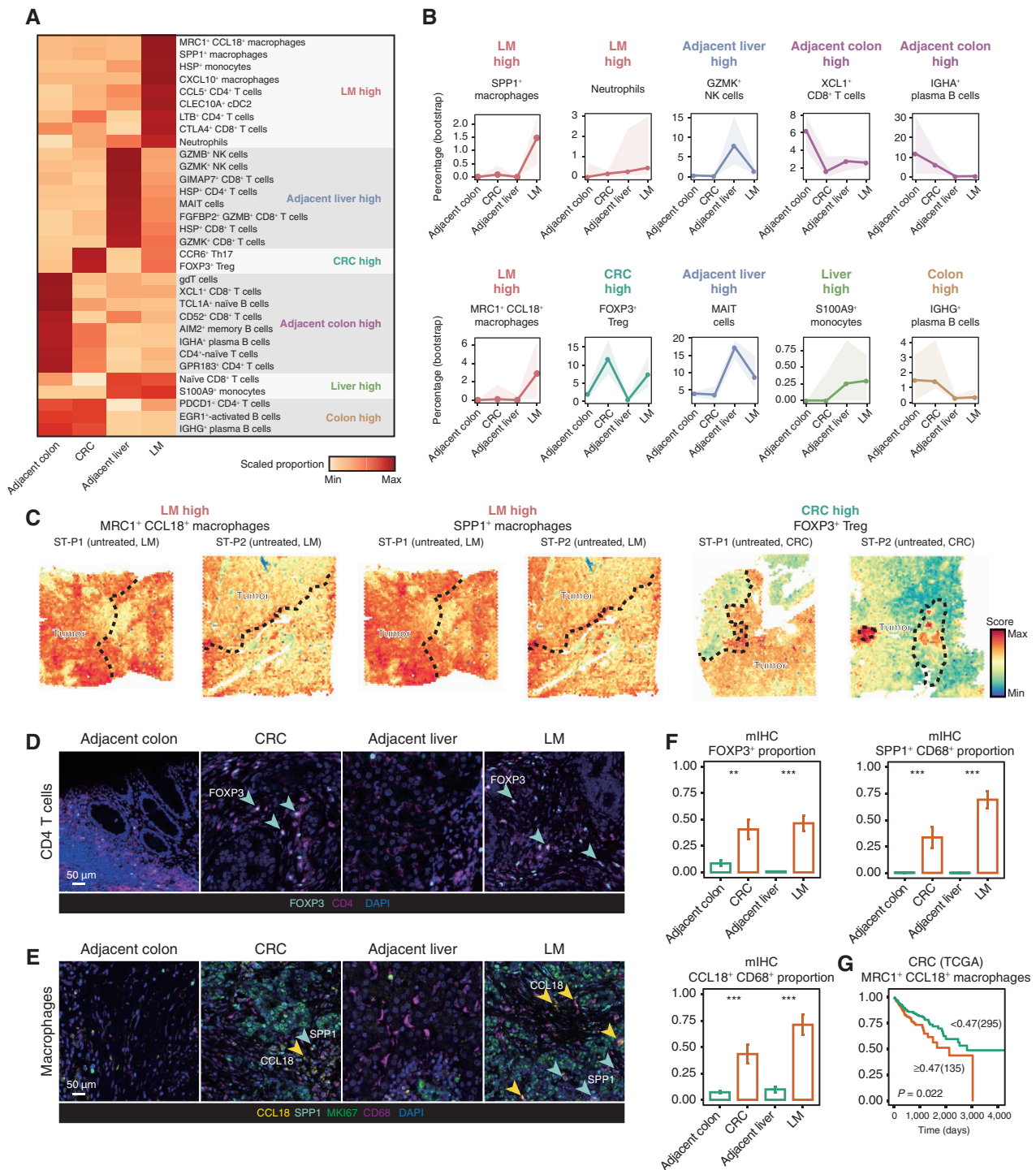
In parallel, ST data of eight samples (paired colorectal cancer and liver metastasis) from two untreated patients and two NAC-PR patients (Fig. 1F and G; Supplementary Fig. S1G and S1H) were analyzed. Unsupervised clustering analysis classified the samples into distinct regions such as tumor and fibroblast regions (Fig. 1F). To integrate the scRNA-seq and ST data, we used Seurat (22) to quantify the main immune subpopulations. In untreated samples, we observed the immune cell pattern consistent with scRNA-seq, such as CD8<sup>+</sup> T cells' remarkable enrichment in both liver metastasis and colorectal cancer (Fig. 1G). By contrast, in NAC-treated samples, we observed smaller tumor regions and irregular distribution of immune cells. Especially in liver metastasis, only a small fraction of spots was clustered into cancer cells, indicating the fundamental remodeling of cellular compartment after NAC treatment. Taken together, the results highlighted the spatiotemporal dynamic immune cell landscape of patients with CRLM and the huge remodeling

after NAC. The single-cell CRLM atlas is now available online for exploring the immune diversity in primary and metastatic sites (<http://www.cancerdiversity.asia/scCRLM>).

### Metastatic Tumors Are Enriched with Immune-Suppressive Cells, Especially *MRC1*<sup>+</sup>*CCL18*<sup>+</sup> M2-like Macrophages

Given the heterogeneous cellular compositions across patients, we utilized a bias-corrected and accelerated (BCa) bootstrap algorithm to estimate the cell proportion that uses bootstrap resampling for 1,000 times to estimate the SEs associated with cell-type proportion estimations (23). We filtered low-abundant immune cells, clustered the immune cell proportion, and scaled them across tissues (Fig. 2A; Supplementary Fig. S2A). In general, immune cell abundance could be classified into six distinct patterns in untreated samples (adjacent colorectal high, colorectal cancer high, adjacent liver high, LM high, colorectal high, and liver high). Adjacent colorectal samples showed the highest proportions of *AIM2*<sup>+</sup> memory B cells, *TCL1A*<sup>+</sup> naïve B cells, and CD4<sup>+</sup> naïve T cells, whereas colorectal cancer samples were mostly enriched with *FOXP3*<sup>+</sup> Tregs, MAIT cells, NK cells, and *FGFBP2*<sup>+</sup> *GZMB*<sup>+</sup> CD8<sup>+</sup> T cells were enriched in adjacent normal liver tissues, analogous to our previous reports in primary liver cancer (24, 25). These results indicated an organ-specific distribution of certain immune cells. Interestingly, *PDCD1*<sup>+</sup> CD4<sup>+</sup> T cells and *CTLA4*<sup>+</sup> CD8<sup>+</sup> T cells were relatively enriched in primary tumors (2.24% and 1.49% of CD45<sup>+</sup> cells) and liver metastases (1.17% and 3.02% of CD45<sup>+</sup> cells), respectively, which is in accordance with other computational predictions (26). The co-occurrence of these cells in the TME indicated a common tumor education on immune cells. In contrast, some immunosuppressive cells were specifically present in liver metastasis, among which a sharp increase of *SPPI*<sup>+</sup> macrophages and *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages was observed (Fig. 2B). We asked the origin of such liver metastasis-specific macrophages and found that especially the *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages shared the Kupffer cell signature (27), indicating their potential origin from hepatic Kupffer cells (Supplementary Fig. S2B; Methods). Neutrophils, which were reported to function as potential protumor players (28), were also enriched in liver metastasis (Fig. 2B). We further used single-sample gene set enrichment analysis (ssGSEA), a method proven to be robust for ST data (29), to quantify the spatial distribution of the immune cell subsets. Again, *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and *SPPI*<sup>+</sup> macrophages were enriched in treatment-naïve liver metastasis (Fig. 2C). Those data were partly consistent with the scRNA-seq data, further confirming the potentially suppressive TME at liver metastasis.

We next sought to validate our findings in another three independent CRLM cohorts. First, we used multiplex IHC (mIHC) to quantify spatial distribution of selected immune cell subsets in additional 27 patients with CRLM ( $n = 18$ , untreated;  $n = 9$ , NAC-PR) in the same hospital (Fig. 2D–F; Supplementary Table S1). Among the untreated samples, we observed the significantly increased infiltration of *FOXP3*<sup>+</sup> Tregs, *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages, and *SPPI*<sup>+</sup> macrophages in colorectal cancer and liver metastasis, largely consistent with our scRNA-seq data (Fig. 2F). Second, we utilized ssGSEA to score the immune cells in a bulk microarray data set of CRLM (ref. 30;  $n = 133$ ; see Methods). Because of lack of



**Figure 2.** Immune remodeling from primary tumor to LM, especially the  $MRC1^+ CCL18^+$  macrophages. **A**, Heat map showing proportions of immune cell subsets in different tissues. The color represents the scaled cellular proportion. The pattern of immune infiltration can be classified into six types, including adjacent colon high, colorectal cancer (CRC) high, adjacent liver high, liver metastasis (LM) high, colon high, and liver high. **B**, The proportions of selected immune cell subsets. The y-axis represents the percentage (bootstrap; Methods), and the x-axis represents different tissues. The shaded areas represent the upper quantile and lower quantile bootstrap cell proportions. **C**, The ST of selected immune cell subsets in colon or liver tumor-normal samples. **D**, Multiplex IHC of  $FOXP3^+$  Tregs. The blue arrows represent the  $FOXP3^+$  Tregs. **E**, Multiplex IHC of  $SPP1^+$  macrophages and  $MRC1^+ CCL18^+$  macrophages. Blue arrows represent the  $SPP1^+$  macrophages, and yellow arrows represent the  $MRC1^+ CCL18^+$  macrophages. **F**, The proportions of  $SPP1^+$  macrophages,  $MRC1^+ CCL18^+$  macrophages, and  $FOXP3^+$  Tregs among all  $CD45^+$  cells revealed by multiplex IHC. Box plot, mean value; error bar, standard error value. \*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.005$ .  $P$  values were determined by the Wilcoxon test. **G**, The prognostic value of  $MRC1^+ CCL18^+$  M2-like macrophages in TCGA patients with colorectal cancer.  $P$  values were determined by the log-rank test.

adjacent tissue data, we could only check the cell abundance between primary and metastatic tumors. In accordance with our scRNA-seq data, *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages, *SPPI*<sup>+</sup> macrophages, and neutrophils were all significantly higher in metastatic tumors in this validation cohort (Supplementary Fig. S2C). Third, we explored the prognostic value of scRNA-seq defined immune cell subpopulations in The Cancer Genome Atlas (TCGA) colorectal cancer cohort (ref. 31; *n* = 430, Fig. 2G; Supplementary Fig. S2D). The high scoring of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and *SPPI*<sup>+</sup> macrophages in primary tumors both predicted a significantly worse prognosis (Fig. 2G; Supplementary Fig. S2D). To summarize, our single-cell and spatial immune landscape of CRLM could be validated in independent cohorts, indicating the robustness of our data. These observations also led us to hypothesize that specific macrophage subpopulations may play fundamental roles in protumor niche formation in CRLM.

### Metastatic Tumor Potentially Educated Macrophages Shift toward Suppressive Status

Given the enrichment of macrophage subpopulations in liver metastasis, we hypothesized that those liver metastasis-enriched macrophages might be functionally distinct across different tissues. We hence performed the ligand-receptor-based cancer-immune cross-talk analysis and compared between primary and metastatic sites (Fig. 3A; Methods). Strikingly, *SPPI*<sup>+</sup> macrophages and *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages ranked high among all differential cross-talking cell types between liver metastasis and colorectal cancer. This result suggested that liver metastatic cancer cells may preferentially reprogram macrophages and induce their specific functional status (Fig. 3A), which is likely due to the intrinsic difference between primary and metastatic cancer cells (Supplementary Fig. S3A). Of note, metastatic tumor cells in liver metastasis showed preferential expression of ligand *CD47*, which is an important “don’t-eat-me” checkpoint (32, 33), and therefore could potentially recruit or activate *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages through the corresponding receptor *SIRPA* (Fig. 3B). We further utilized mIHC to visualize those proteins and directly observed the cross-talk between *CD47*<sup>+</sup> tumor cells and *SIRPA*<sup>+</sup> macrophages (Fig. 3C). These ligand-receptor pairs specifically enriched in liver metastasis provided clues for targeted therapy against liver metastasis.

To pinpoint the difference imprinted by unique TME, we performed differential gene-expression analysis between colorectal cancer and liver metastasis macrophages. Notably, liver metastasis-enriched *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages highly expressed a broad spectrum of key molecules fundamental for macrophage polarization (Fig. 3D; Supplementary Table S3). *APOE*, which functions as an anti-inflammatory and pro-M2 conversion protein (34), was one of the significantly differentially expressed genes. Other M2 polarization-related genes such as *MARCO* (35) were also significantly upregulated in liver metastasis-enriched *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages (Fig. 3E). By contrast, we found colorectal cancer-enriched *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages showed higher expression of inflammatory cytokines including *TNF*, *IL1B*, *CCL3*, and *CCL4*. Such shift of molecular profiles was also observed in *SPPI*<sup>+</sup> macrophages, indicating that macrophages shared common functional changes from primary to metastatic

sites (Supplementary Fig. S3B and S3C; Supplementary Table S3). Then, pathway enrichment analyses showed (Fig. 3F) a strong enrichment of metabolic pathways in *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages from both colorectal cancer and liver metastasis, highlighting the fundamental role of metabolism in determining those macrophage functions. Notably, we found that *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages in colorectal cancer were enriched with oxidative phosphorylation, whereas their counterparts in liver metastasis were dominated by amino acid metabolism. These data highlighted that metabolic regulation might mediate the phenotypic and functional shift of macrophages in response to distinct microenvironmental cues.

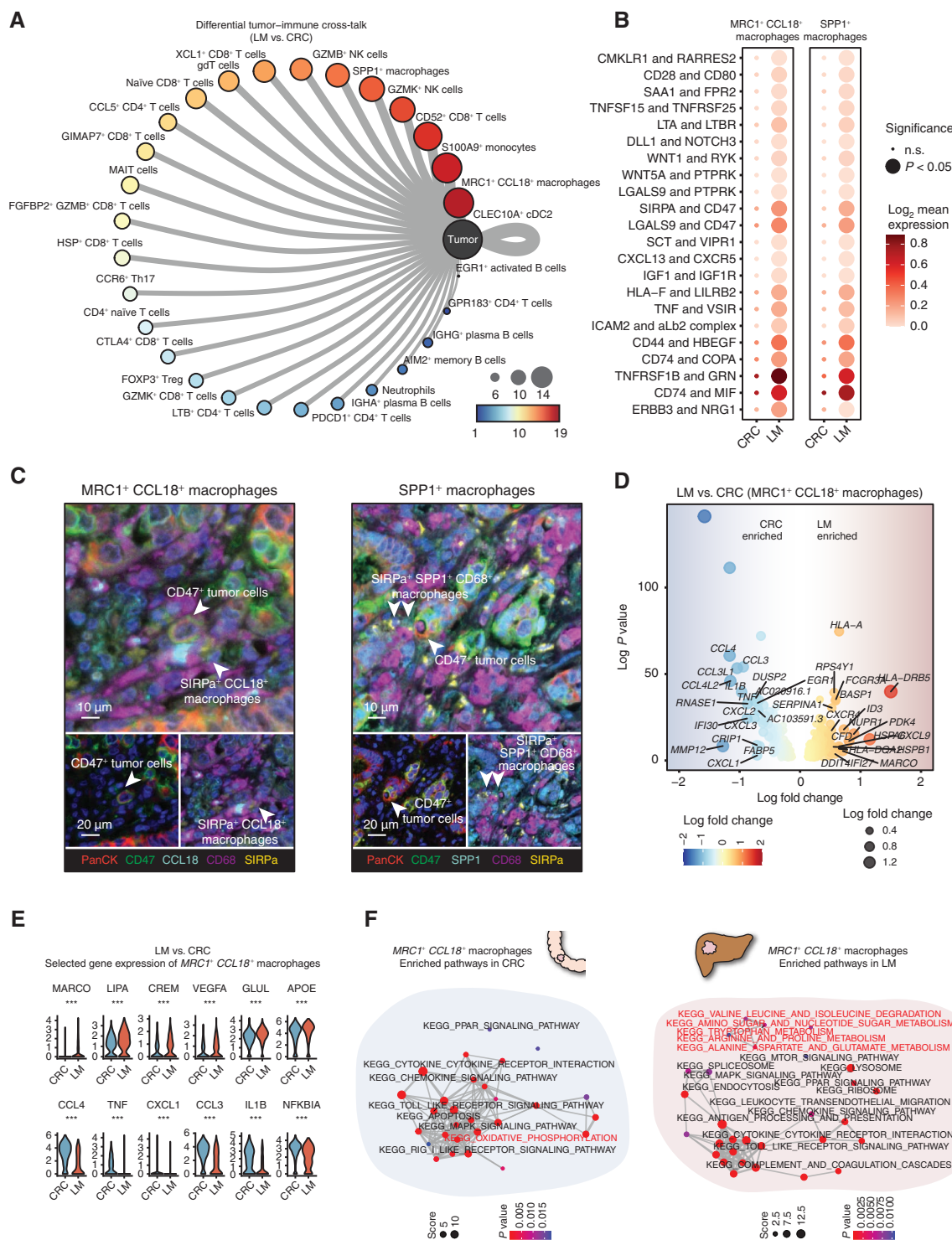
### scMetabolism: A Computational Pipeline for Quantifying Metabolic Activity Using scRNA-seq Data

Mining metabolic activity at single-cell resolution remains challenging, mostly due to lack of an easy-to-use computational pipeline. Although Locasale Lab recently reported their ssGSEA-based workflow (36), performing such analysis still requires a strong computational background and is time-consuming. Here, we present an R package, scMetabolism, which integrates dropout gene imputation, metabolism quantification, and data visualization. We demonstrated that this workflow shows good compatibility with Seurat scRNA-seq toolkit (22, 37) and is broadly applicable to scRNA-seq data sets. The R package of scMetabolism is now available on Github (<https://github.com/wu-yc/scMetabolism>).

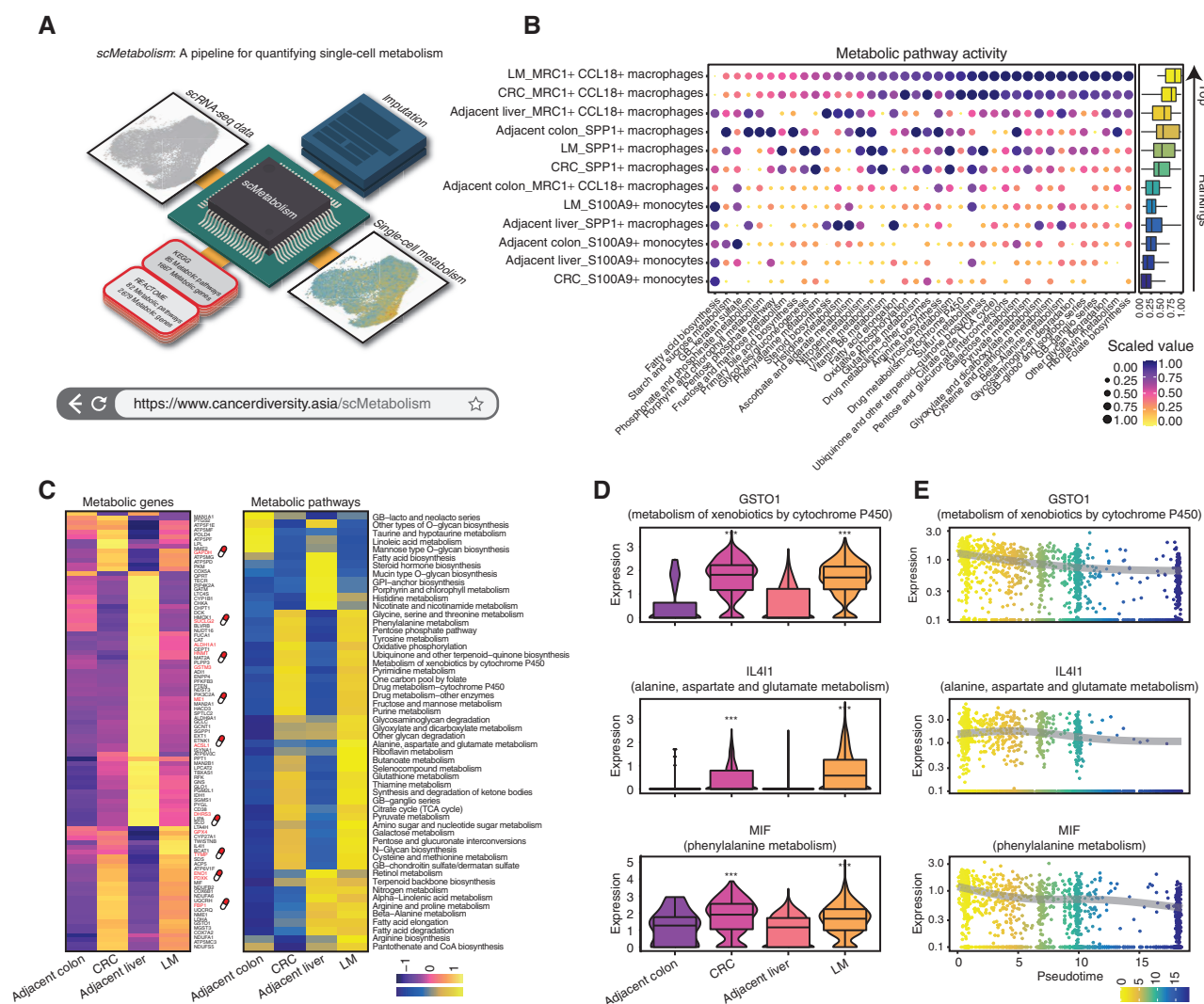
The scMetabolism package consists of three serial steps (Fig. 4A). First, scMetabolism supports the imputation of the dropout value based on ALRA (38). Second, scMetabolism quantifies the metabolic pathway activity based on VISION (39), AUCell (40), and ssGSEA (41). We integrated reported gene sets (36) and manually reviewed gene sets from KEGG (42, 43) and REACTOME (43, 44) to generate a list of high-quality metabolic gene sets. Third, scMetabolism provides three different ways, including box plot, uniform manifold approximation and projection (UMAP)/t-distributed stochastic neighbor embedding plot, and dot plot, to visualize the metabolic diversity of single cells (see Methods). We also tested the accuracy (comparison with published methods; ref. 36), consistency (correlation between different methods such as VISION and AUCell), and stability (dropout simulation) of scMetabolism on four different data sets, including PBMCs, melanoma, head and neck squamous cell carcinoma, and testis (45–47), to test the performance of this pipeline (Supplementary Fig. S4A–S4E; see Methods). Finally, we developed an easy-to-use web server, scMetabolism online (<http://www.cancerdiversity.asia/scMetabolism/>), for biologists to easily quantify metabolic activity of single cells.

### A Sharp Increase of Metabolic Activity of *MRC1*<sup>+</sup>*CCL18*<sup>+</sup> Macrophages in Liver Metastasis

To understand the metabolic landscape of myeloid cells in CRLM, we first computed the score of all 76 active metabolic pathways in suppressive macrophages and monocytes. Among them, we selected top 50% variable metabolic scores and ranked myeloid cells (excluding myeloid cells with low abundance) according to their average metabolic score (Fig. 4B). Strikingly, liver metastasis-infiltrated *MRC1*<sup>+</sup> *CCL18*<sup>+</sup>



**Figure 3.** MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages in metastatic tumors exhibited terminally differentiated and suppressive states. **A**, The ranked differential tumor-immune cell cross-talk [liver metastasis (LM) vs. colorectal cancer (CRC)] shows MRC1<sup>+</sup> CCL18<sup>+</sup> M2-like macrophages ranked the second among all ligand-receptor pairs. **B**, The liver metastasis upregulated ligand-receptor cross-talk between MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages and SPP1<sup>+</sup> macrophages with cancer cells (liver metastasis vs. colorectal cancer). The y-axis represents the ligand and receptor name, whereas the x-axis represents the tissue. The circle size represents the log-normalized P value, whereas the color darkness represents the log-transformed mean expression of ligand and receptor. **C**, Multiplex IHC shows the cross-talk between tumor cells and MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages or SPP1<sup>+</sup> macrophages via the ligand-receptor of CD47-SIRPA. **D**, The volcano plot represents the differentially expressed genes of MRC1<sup>+</sup> CCL18<sup>+</sup> M2-like macrophages between colorectal cancer and LM. **E**, The selected gene expression of MRC1<sup>+</sup> CCL18<sup>+</sup> M2-like macrophages in colorectal cancer and LM. \*\*\*,  $P < 0.005$ . P values were determined by the Wilcoxon test. **F**, Pathway enrichment analysis of highly expressed genes of MRC1<sup>+</sup> CCL18<sup>+</sup> M2-like macrophages in colorectal cancer and liver metastasis, respectively. KEGG gene sets were used to perform the pathway enrichment analysis (Methods). Only selected pathways are shown.



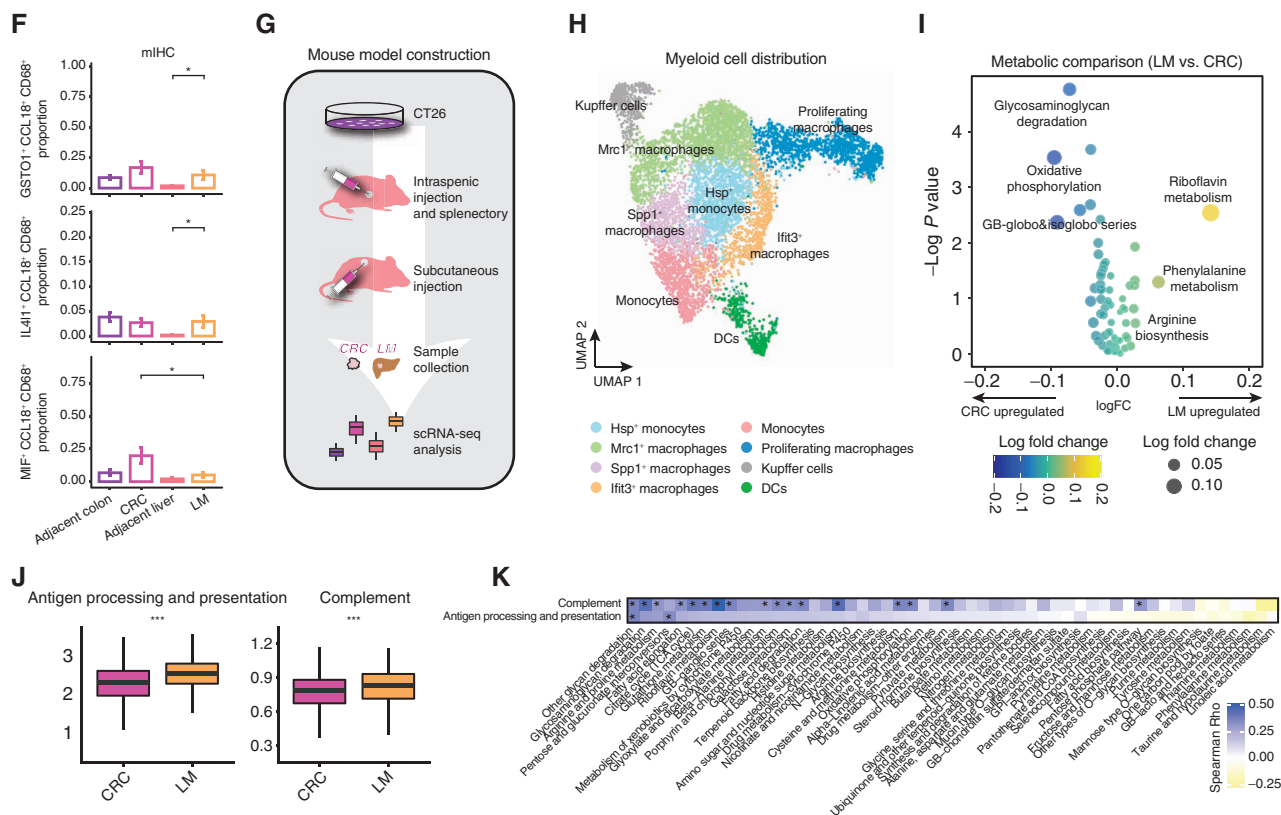
**Figure 4.** *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages in metastatic tumors showed high metabolic activity. **A**, The schematic plot of *scMetabolism*, a pipeline for quantifying single-cell metabolism based on scRNA-seq. **B**, The metabolic activity analysis of myeloid cells shows that *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and *SPP1*<sup>+</sup> macrophages in metastatic tumors have the highest metabolic score. The circle size and color darkness both represent the scaled metabolic score. **C**, The heat map of average metabolic gene expression and median metabolic pathway score of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages. The genes highlighted with red represent the potential druggable genes (Methods). **D**, The expression of selected metabolic genes in adjacent colon, colorectal cancer, adjacent colon, and LM. \*\*\*,  $P < 0.005$ .  $P$  values were determined by the Wilcoxon test between tumor and their adjacent tissues. **E**, The correlation between pseudotime and metabolic gene expression shows the liver metastasis macrophages highly express those metabolic genes. (continued on next page)

macrophages showed the highest metabolic activities among all myeloid cells, indicating that they were extremely energetic, which could be linked to their unique functions at metastatic sites.

Further unsupervised clustering of all metabolism-associated genes (see Methods) of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages observed a strong metabolic preference associated with surrounding tissues (Fig. 4C), identifying 42 metabolism pathways potentially upregulated in colorectal cancer and liver metastasis-infiltrated *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages. Among them, phenylalanine metabolism, which aids in generating tyrosine (48), was more highly expressed in liver metastasis-enriched *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages. We subsequently closely looked at the differentially expressed metabolic genes

and found metabolic genes specifically upregulated in primary and metastatic infiltrated *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages, which were partly the target genes of known drugs (fetched from Enrichr; Fig. 4C; Methods; ref. 49). *GSTO1*, an enzyme in metabolism of xenobiotics by cytochrome P450 (50), was significantly higher in colorectal cancer and liver metastasis-infiltrated *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and associated with differentiation (Fig. 4D and E). *IL4I1* and *MIF*, genes essential for macrophage activation (51–53), also showed similar features (Fig. 4D and E). We validated those metabolic enzymes by using mIHC and observed largely consistent results (Fig. 4F; Supplementary Fig. S4F and S4G).

We next asked whether such highly metabolic activated states of macrophages were conserved across species. We

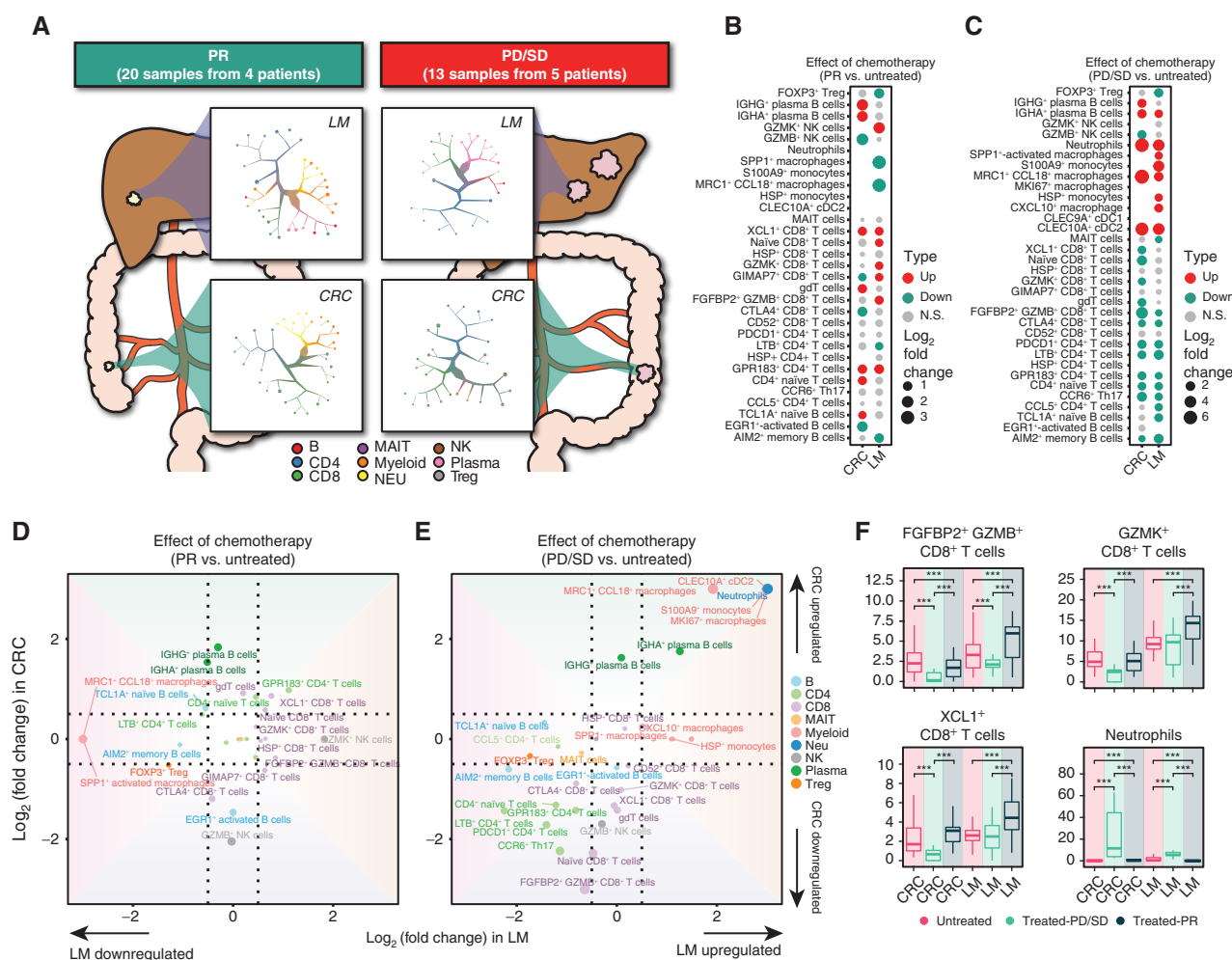


**Figure 4. (Continued)** **F**, Multiplex IHC shows the proportions of *CCL18*<sup>+</sup> macrophages that harbor high metabolic enzyme expression (GSTO1, IL411, and MIF). The y-axis represents the proportion of triple-positive cells of all *CD68*<sup>+</sup> cells. *P* values were determined by the Wilcoxon test. **G**, The workflow of mouse model construction. **H**, The UMAP plot of all macrophages and monocytes of mouse CRLM scRNA-seq data. **I**, The volcano plot represents the differentially metabolic activity of *Mrc1*<sup>+</sup> macrophages between colorectal cancer and liver metastasis. *P* values were determined by the Wilcoxon test. **J**, The phenotypic difference of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages between colorectal cancer and liver metastasis. *P* values were determined by the Wilcoxon test. **K**, Correlation between phenotypic score and metabolic activity score in liver metastasis-infiltrated *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages. Only tumor high metabolic pathways were used for analysis. \*, *P* < 0.05 and Spearman Rho > 0.3. *P* values were determined by the Spearman rank test.

hence constructed the mouse model of CRLM (Fig. 4G; Methods) and applied scRNA-seq on the primary/metastatic tumors (Supplementary Fig. S4H and S4I). As a result, similar macrophage subpopulations, such as *Mrc1*<sup>+</sup> macrophages and *Spp1*<sup>+</sup> macrophages, were identified in the mouse CRLM model (Fig. 4H). However, we did not observe the significantly differential proportion of these macrophages between primary and metastatic sites, possibly due to the fact that primary tumors were established subcutaneously rather than in the colon (Supplementary Fig. S4J). Further metabolic quantification showed the featured metabolic states enriched in liver metastasis and colorectal cancer-infiltrated macrophages (Fig. 4I). We checked the top three differentially expressed pathways (liver metastasis vs. colorectal cancer in the mouse model), and some of them were conserved in human and mouse. For example, oxidative phosphorylation and phenylalanine metabolism were in consistency between human and mouse macrophages. The expression of MIF, the key enzyme of phenylalanine metabolism, is also consistent with human CRLM (Supplementary Fig. S4K). In summary, macrophages in liver metastasis harbored a sharp increase of metabolic activity such as phenylalanine metabolism. Inhibiting such metabolism activity might mobilize an antitumor

immune response to control hepatic metastases in advanced colorectal cancer.

To illustrate the association of metabolic reprogramming with phenotypic changes of macrophages, we computed the classic phenotypic scores of macrophages such as antigen processing and presentation, inflammation, angiogenesis, immune suppression, phagocytosis, interleukin signaling pathway, and complement (Fig. 4J; Supplementary Fig. S4L and S4M; Supplementary Table S4). Indeed, within liver metastasis, *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages harbored significantly increased scores of antigen processing and presentation and complement activity (Fig. 4J; Supplementary Fig. S4N), and *SPPI*<sup>+</sup> macrophages also showed similar trend (Supplementary Fig. S4L). Certain genes like HLA-A (Supplementary Fig. S4M), one of the MHC class I molecules (54), were upregulated in liver metastasis-enriched *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages. We further analyzed the correlation between the 42 tumor-high metabolic pathways and phenotypic scores of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages. Interestingly, several macrophage-specific metabolic pathways, including pentose and glucuronate interconversions, showed strong correlations with antigen processing and presentation in liver metastasis but weak in colorectal cancer (Fig. 4K; Supplementary Fig. S4O). These data collectively suggested the potentially



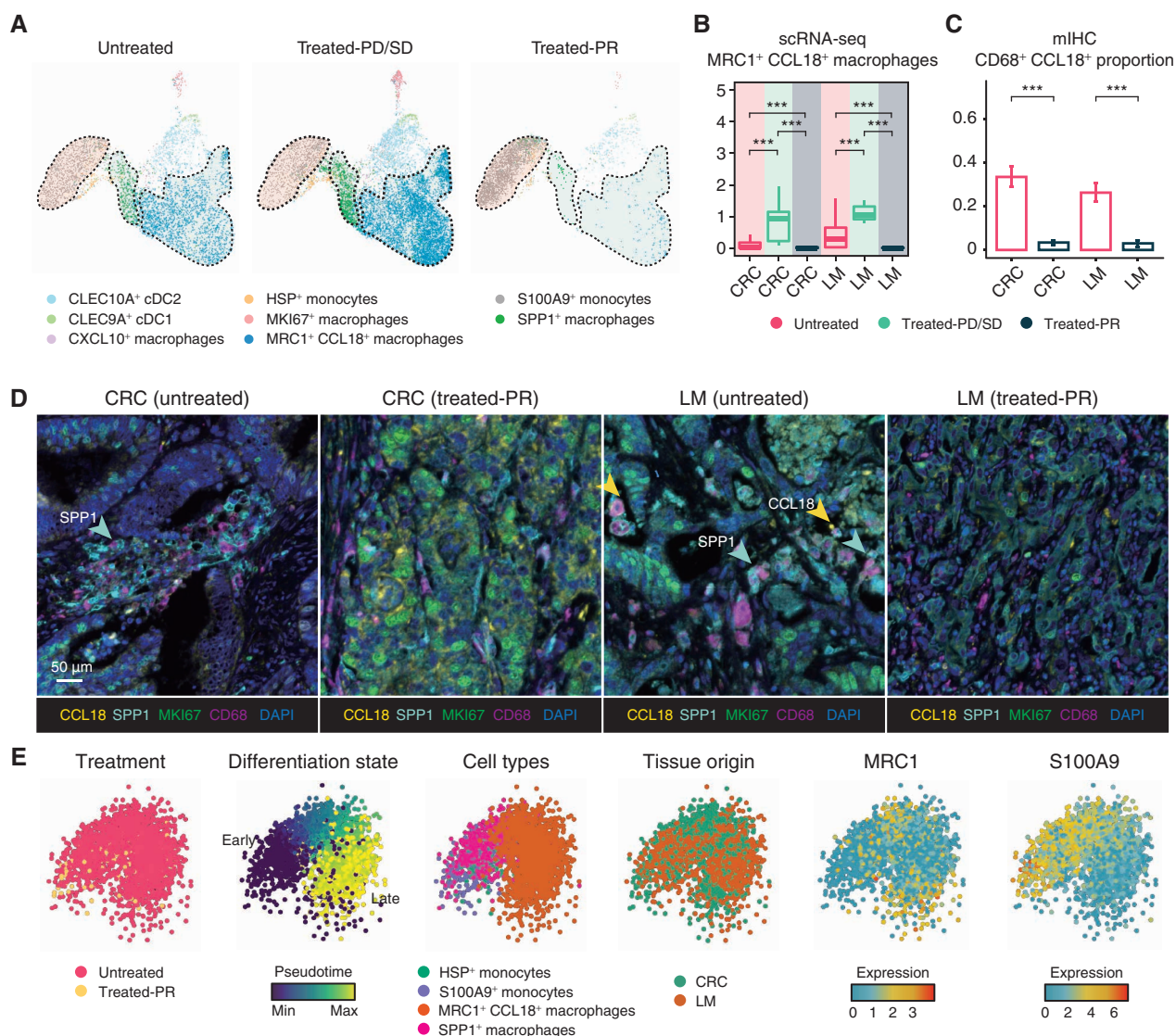
**Figure 5.** Neoadjuvant chemotherapy restored the immune balance by activating cytotoxic immune cells and inhibiting immunosuppressive cells in metastatic tumors. **A**, The cell clustering in chemotherapy PD/SD, and PR tumors are largely different. **B**, The dot plot shows the cell composition difference between untreated and NAC-PR tumors. *P* values were determined by the Wilcoxon test of bootstrap cellular proportion. **C**, The dot plot shows cell composition difference between untreated and NAC-PD/SD tumors. *P* values were determined by the Wilcoxon test of bootstrap cellular proportion. **D**, The cellular changes between untreated and NAC-PR tumors. *P* values were determined by the Wilcoxon test of the bootstrap cellular proportion. **E**, Cellular changes between untreated and NAC-PD/SD tumors. *P* values were determined by the Wilcoxon test of the bootstrap cellular proportion. **F**, Proportional changes of selected cell types between untreated, PR, and PD/SD samples. \*\*\*, *P*<sub>adj</sub> < 0.005. Adjusted *P* values were determined by the Wilcoxon test.

essential role of metabolism in determining *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages' phenotypes.

### NAC-Responsive and Nonresponsive Tumors Harbor Distinct Immune Dynamics, Especially Macrophages

To explore the effect of NAC on antitumor immunity, we performed the clustering analysis of all CD45<sup>+</sup> cells using TooManyCells (55). Interestingly, we observed strong difference of cell clustering between colorectal cancer and liver metastasis (Fig. 5A; Supplementary Fig. S5A). To understand which cell types were changed, we computed the bootstrapped cell proportion and performed the differential analysis (Fig. 5B and C; see Methods). Of note, the impact of NAC on the TME was highly cell type-specific and context-dependent (Fig. 5B and C). Generally, in PR tumors, NAC could downregulate immunosuppressive cells and activate cytotoxic cells

in tumors (Fig. 5B). In liver metastasis, responsive NAC downregulated the *SPPI*<sup>+</sup> macrophages but upregulated cytotoxic T cells such as *GZMK*<sup>+</sup> CD8<sup>+</sup> T cells and *XCL1*<sup>+</sup> CD8<sup>+</sup> T cells (Fig. 5B and D). Particularly, several *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and dozens of *SPPI*<sup>+</sup> macrophages were left in NAC-PR tumors, suggesting a potential role of NAC in inhibiting macrophage lineages. However, in PD/SD tumors, we observed distinct immune cell changes from PR tumors. For example, an increased level of *SPPI*<sup>+</sup> macrophages and *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages was infiltrated in liver metastasis, whereas the cytotoxic immune cells (i.e., *FGFBP2*<sup>+</sup> *GZMB*<sup>+</sup> CD8<sup>+</sup> T cells) were downregulated (Fig. 5C, E, and F). Those opposite results might be explained by the organ-specific and treatment efficacy-related remodeling of cancer cells. For example, metastatic PD/SD cancer cells showed the specific enrichment of DNA repair and fatty acid metabolism, whereas metastatic PR cancer cells were more associated



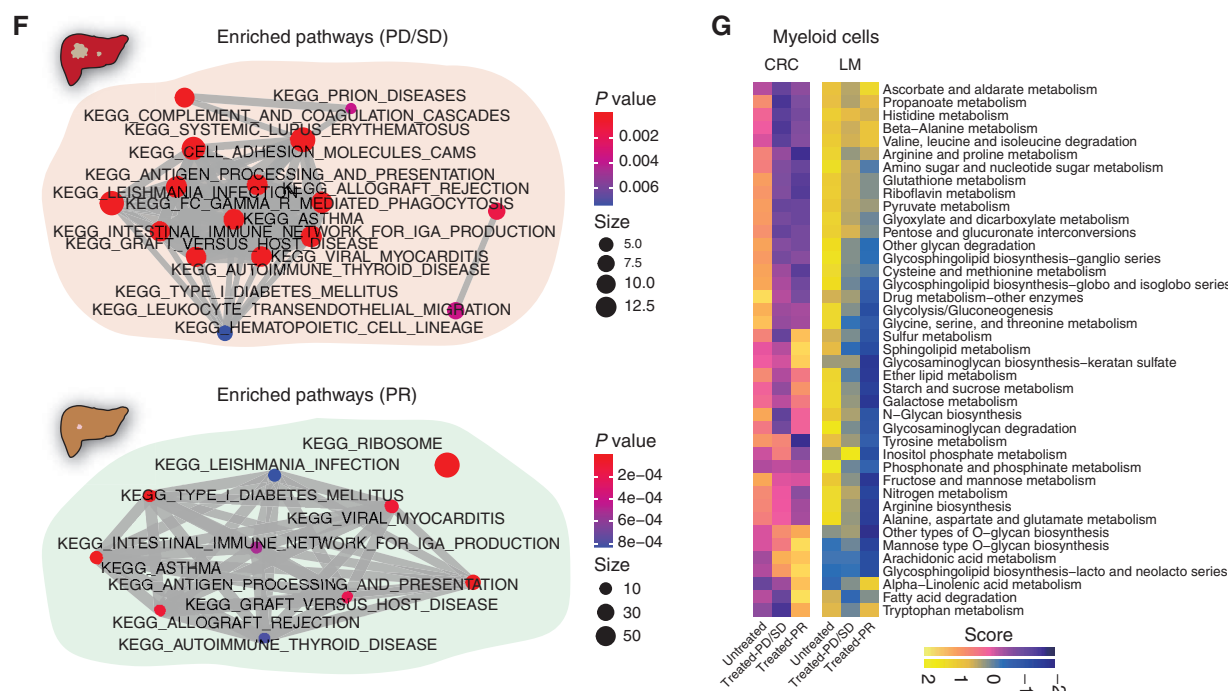
**Figure 6.** Metabolic reprogramming of MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages in neoadjuvant chemotherapy-treated samples. **A**, A representative UMAP map showing the myeloid cell distribution between untreated, NAC-PD/SD, and NAC-PR samples. **B**, Cellular proportional changes of MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages in untreated, NAC-PD/SD, and NAC-PR samples revealed by scRNA-seq. The y-axis represents the proportion of MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages among CD45<sup>+</sup> cells. \*\*\*,  $P < 0.005$ .  $P$  values were determined by the Wilcoxon test. **C**, Cellular proportional changes of MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages in untreated and NAC-treated samples revealed by multiplex IHC. The y-axis represents the proportion of CD68<sup>+</sup> CCL18<sup>+</sup> macrophages among CD68<sup>+</sup> cells. Box plot, mean value; error bar, standard error value. \*\*\*,  $P < 0.005$ .  $P$  values were determined by the Wilcoxon test. **D**, mIHC of SPP1<sup>+</sup> macrophages, MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages, and MKI67<sup>+</sup> macrophages in untreated/treated PR colorectal cancer and liver metastasis samples. The blue arrows represent the SPP1<sup>+</sup> macrophages, and the yellow arrows represent the MRC1<sup>+</sup> CCL18<sup>+</sup> macrophages. **E**, Trajectory plot shows the treatment status, pseudotime, cell-type, tissue, MRC1 expression, and S100A9 expression of all myeloid cells in untreated and treated PR groups. (continued on following page)

with adipogenesis (Supplementary Fig. S5B). Alternatively, NAC treatment may also directly and differentially regulate individual immune subsets. In addition to tumors, NAC upregulated the proportion of GZMB<sup>+</sup> CD8<sup>+</sup> T cells in adjacent normal liver and increased the proportion of FGFBP2<sup>+</sup> GZMB<sup>+</sup> CD8<sup>+</sup> T cells in PBMC (Supplementary Fig. S5C and S5D) of PR patients. As for PD/SD patients, a reduced level of FGFBP2<sup>+</sup> GZMB<sup>+</sup> CD8<sup>+</sup> T cells was also observed in adjacent colon and liver (Supplementary Fig. S5D). Altogether, the results demonstrated that effective NAC not only reprogrammed intratumor immune balance but also activated

systemic antitumor immune responses, providing evidence to support the potential use of NAC in resectable CRLM.

### Metabolic Reprogramming of Macrophages Induced by NAC in Responsive Tumors

Considering that NAC mostly affected myeloid populations, especially macrophages, we hence asked whether the trajectory of macrophages was edited by NAC. We split the UMAP distribution of myeloid cells according to receiving NAC treatment response and identified a rich diversity of myeloid phenotypes between PD/SD and PR samples (Fig. 6A).



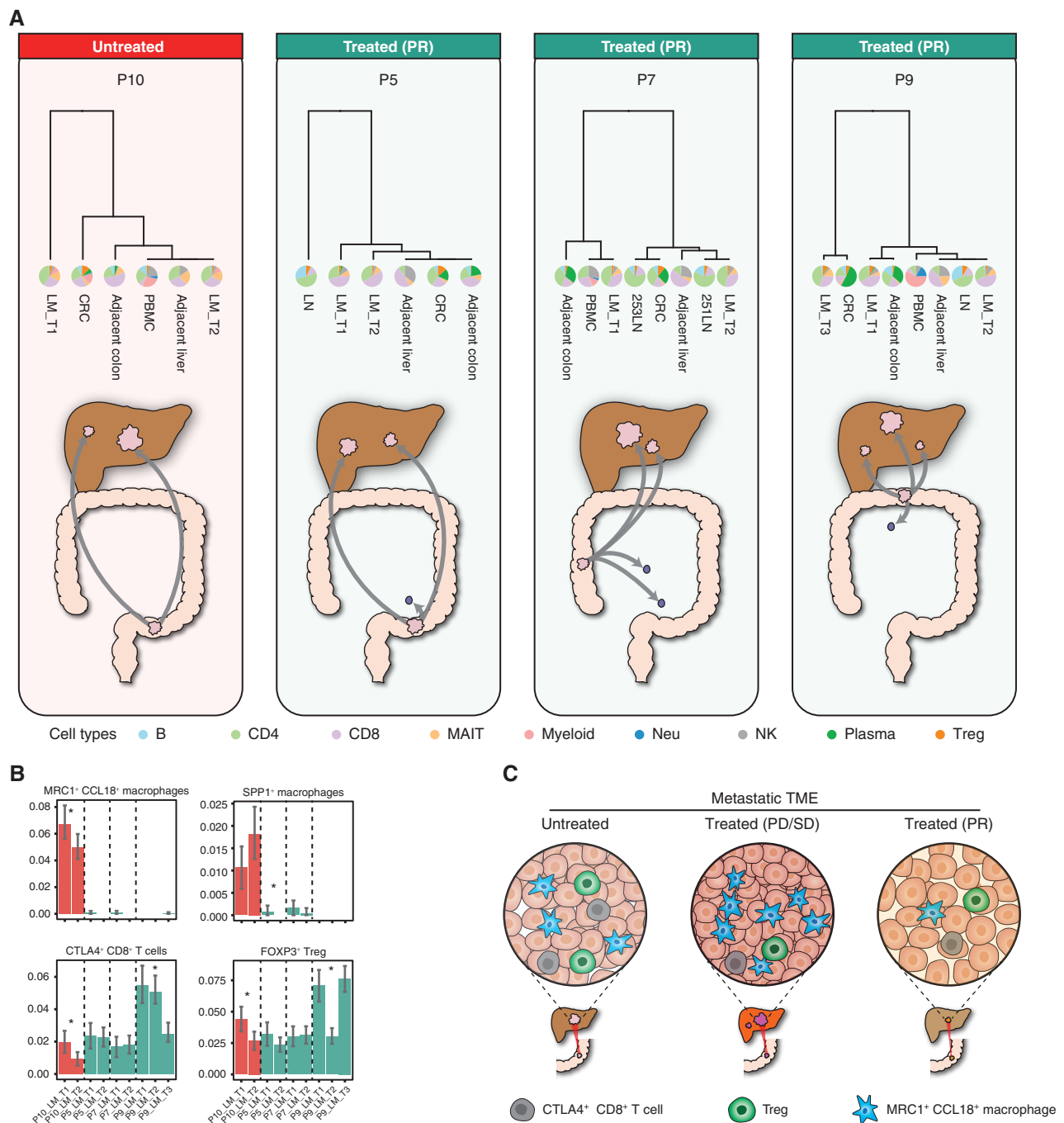
**Figure 6. (Continued)** F, Pathway enrichment analysis of highly expressed genes of myeloid cells in liver metastasis (untreated vs. PD/SD and untreated vs. PR, respectively). KEGG gene sets were used to perform the pathway enrichment analysis (Methods). G, The heat map of significantly altered metabolic pathways of all myeloid cells upon NAC treatment.

In untreated samples, we could observe the developing trajectory of myeloid cells, whereas myeloid cells in the PR group were enriched with a monocyte phenotype, a progenitor state of macrophages (S6), indicating an early differentiation status (Fig. 6A). The bootstrap cell proportion differential analysis confirmed the lack of terminal status of macrophages in NAC-PR tumors, such as a significant downregulation of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and *SPPI*<sup>+</sup> macrophages (Fig. 6B; Supplementary Fig. S6A). By contrast, the NAC-PD/SD tumors harbored a significantly increased level of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and *SPPI*<sup>+</sup> macrophages compared with treatment-naïve tumors (Fig. 6B; Supplementary Fig. S6A). We next utilized mIHC to check those macrophages in another independent cohort ( $n = 18$ , untreated;  $n = 9$ , NAC-PR; Fig. 6C; Supplementary Table S1), confirming the lower infiltration of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and *SPPI*<sup>+</sup> macrophages in NAC-PR liver metastases (Fig. 6D; Supplementary Fig. S6B). Trajectory inferring analysis using dynverse (S7) further identified a clear developing trajectory from *S100A9*<sup>+</sup> monocytes to *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages, indicating the opposite polarization of untreated versus NAC-PR cells (Fig. 6E). By focusing on *MRC1* and *S100A9* expression, which represents *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> macrophages and monocytes, respectively, we again demonstrated that the chemotherapy-responsive cells exhibited monocyte-enriched phenotypes. On the contrary, in the PD/SD group, the trajectory reprogramming of macrophages was not observed (Supplementary Fig. S6C). The results indicate that the trajectory blocking of macrophages by chemotherapy is exclusive in responsive tumors.

Given the limited number of *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> and *SPPI*<sup>+</sup> macrophages in metastatic tumors, we performed the pathway enrichment analysis of all myeloid cells between untreated and NAC-treated samples and found that NAC induced specific functional changes of myeloid cells in the PR and PD/SD groups (Fig. 6F; Supplementary Fig. S6D). Liver metastasis macrophages in the PR group showed the enrichment of ribosome gene signature, whereas those in the PD/SD group were associated with complement and coagulation cascades. Then, the scMetabolism pipeline, focusing on metabolic changes, showed that NAC considerably blocked myeloid metabolism and the PR group showed more downregulated metabolism (Fig. 6G). The glycolysis and gluconeogenesis were consistently downregulated in macrophages from the PD/SD and PR groups in both colorectal cancer and liver metastasis. By contrast, macrophages in the PD/SD group were also associated with specific metabolic pathways such as inositol phosphate metabolism upregulation in colorectal cancer and liver metastasis. Together, NAC exerted specific metabolic reprogramming of myeloid lineage, which may extend our understanding of the NAC-induced antitumor mechanisms and informing potential treatment for patients with resectable CRLM.

### Immune Microenvironment of Multifocal Metastasis Is Highly Diverse

A significant proportion of tumors within an organ are multifocal tumors, which harbor diverse genetic, molecular, and immunologic features (S8–S62), as demonstrated by the recently reported heterogeneous TME of multifocal primary



**Figure 7.** The heterogeneous immune microenvironment of multifocal metastasis. **A**, The clustering of all immune cells in four patients with multifocal metastases (P10, P5, P7, and P9). **B**, The proportion of immunosuppressive cells is heterogeneous across multifocal metastases. \*,  $P < 0.05$ .  $P$  values were determined by the Wilcoxon test. **C**, Schematic diagram depicting the unique dynamics of protumor niche in CRLM upon neoadjuvant chemotherapy.

liver cancer (60, 63, 64). Whether multifocal liver metastasis harbors a diverse immune microenvironment and distinct response to chemotherapy remains largely unclear. To generally capture intermetastatic immune heterogeneity, we performed the unsupervised clustering of all CD45<sup>+</sup> cells of four patients with multifocal liver metastasis (P5, P7, P9, and P10). Interestingly, multifocal metastases showed distinct immune

profiles (Fig. 7A). As for the untreated patient (P10), liver metastasis site 2 was in the same cluster of adjacent liver, whereas liver metastasis site 1 was not similar to any other samples. Another three NAC-treated multifocal liver metastases were also featured with intermetastatic heterogeneous immune profiles. Next, bootstrap cell proportion analysis showed that, in untreated samples, immunosuppressive cells

were significantly and differentially infiltrated between different liver metastasis sites, where the *MRC1<sup>+</sup> CCL18<sup>+</sup>* macrophages were differentially infiltrated in liver metastasis of P10 (Fig. 7B). They also harbored distinct intermetastatic phenotypic/metabolic activities such as antigen processing and presentation (Supplementary Fig. S7A and S7B). Thus, such considerable intermetastatic heterogeneity may result in differential responses to chemotherapy and other treatments, which is well documented in clinical practice (65, 66).

## DISCUSSION

Here, we defined the cellular and spatial immune landscape of CRLM by using the state-of-the-art high-throughput scRNA-seq and ST. It is known that the genomic divergence between the primary and metastatic colorectal cancer cells is relatively low (67–69). However, our data revealed that the immune microenvironment has undergone extensive spatiotemporal remodeling in liver metastasis with a strong enrichment of immunosuppressive cells (Fig. 7C), consistent with previous bulk sample-based studies (5, 26, 70). In particular, *MRC1<sup>+</sup> CCL18<sup>+</sup>* macrophages showed a terminally differentiated state, exhibited a metabolically energetic phenotype, and were sensitive to neoadjuvant chemotherapy. Our study not only unravels a previously unrecognized diversity of CRLM immune ecosystem, but also provides a comprehensive resource for the cancer research community.

By applying scMetabolism, a novel R package for quantifying the single-cell metabolic activity on our CRLM data, we surprisingly found that the suppressive *MRC1<sup>+</sup> CCL18<sup>+</sup>* macrophages displayed the highest metabolic activity, especially in the metastatic tumors, suggesting that intratumor immunosuppressive cells were highly energetic. However, after chemotherapy, metabolism of such immunosuppressive cells was significantly slowed down. For example, alanine, aspartate, and glutamate metabolism of *MRC1<sup>+</sup> CCL18<sup>+</sup>* macrophage cells, which is essential for macrophage development and fate decision (71), was downregulated by NAC. Our scMetabolism pipeline not only reveals the metabolic landscape of CRLM immune microenvironment, but also opens new computational opportunities for discovering functional metabolic checkpoints and pathways for other scRNA-seq data sets.

Compared with NAC-associated immune activation (increased infiltration of NK cells and oligoclonal expansion of T cells) in advanced ovarian cancer (72), we surprisingly observed that the cellular states of macrophages were reprogrammed in NAC-responsive patients. In PR samples, there existed only a small fraction of terminally suppressive macrophages such as *MRC1<sup>+</sup> CCL18<sup>+</sup>* macrophages and *SPPI<sup>+</sup>* macrophages, whereas an opposite trend was observed in PD/SD samples. Mechanically, one possibility is that such immune reprogramming was induced by the chemotherapeutic modulation of cancer cells, and another group has also reported the chemomodulation effect by utilizing an *ex vivo* culture platform (73). Accompanied with the reduction of immunosuppressive cells, significant elevation of cytotoxic cells, such as *XCL1<sup>+</sup> CD8<sup>+</sup>* T cells, was noted. Therefore, our data showed that NAC powerfully restored intratumor immune balance in chemotherapy-responsive patients,

decoding additional mechanisms that favored neoadjuvant chemotherapy in those patients.

Metastatic multifocal tumors harbored diverse immune microenvironmental composition. Even inside the same tissue of the same patient, the suppressive immune cells were characterized by differential abundance and distinct functional states. As for multifocal hepatocellular carcinoma, we recently reported that distinct lesions induced heterogeneous T-cell receptor expansions, which may impose immune selection (63). Another independent group also found that, in a patient with high-grade serous ovarian cancer with multifocal lesions (62), the progressive metastases excluded immune cells whereas the regressive lesions were infiltrated by CD8 T cells. This could partly explain the diverse drug responses of multifocal tumors of the same patients in the clinical practice. Our data will allow novel design of optimal rational combination therapies for multifocal CRLM such as combining NAC, metabolism checkpoint inhibitors, and immunotherapy.

Our study did have some limitations. Due to the limited patient number fitting the inclusion criteria (chemotherapy regimen and sample resectability), only four patients were recruited in our ST cohort. Second, the ST technique has not reached the single-cell resolution. Current 10X Visium platform merely supported an estimate of 1 to 10 cells in each spot (74). A more precise spatial map, such as Slide-seq and DBiT-seq (75–77), will be essential for better understanding how to therapeutically target the spatiotemporal heterogeneity of this complex disease. Third, it still remains unclear how the chemotherapy induces the functional changes of macrophages in metastatic tumors. Further experimental validation is required to validate that such state shift of macrophages is due to altered differentiation or population change (i.e., macrophage depletion or monocyte recruitment).

In summary, we present the spatial and cellular atlas of CRLM and pinpoint the functional impacts of neoadjuvant chemotherapy on TME. Our scMetabolism pipeline not only deepens our understanding into the immunobiology of CRLM, but also enables the unbiased discovery of active metabolic states in other complex tissues. These findings provide unique insights into the biology of cancer spreading and raise the possibility to target metabolism pathways in metastasis.

## METHODS

### Sample Characteristics

We collected samples of pathologically diagnosed CRLM from Zhongshan Hospital, Fudan University, with written informed consent and approval from the Institutional Review Board–approved protocols (B2020-348R). The study was conducted in accordance with ethical guidelines (Declaration of Helsinki). In detail, whether the tumors are resectable has all passed the MDT discussion composed by senior clinicians. In detail, 89 samples from 20 patients ( $n = 11$ , untreated;  $n = 5$ , PD/SD;  $n = 4$ , PR) underwent scRNA-seq, and eight samples from four patients ( $n = 2$ , untreated;  $n = 2$ , PR) were sequenced by ST. Another 104 samples from 27 patients were analyzed by mIHC ( $n = 17$ , untreated colorectal cancer and adjacent colon;  $n = 18$ , untreated liver metastasis and adjacent liver;  $n = 9$ , treated PR colorectal cancer and adjacent colon;  $n = 8$ , treated PR liver metastasis and adjacent liver).

## Mouse Model Construction

Six-week-old female BALB/c mice were ordered from Shanghai Model Organisms Center maintained under specific pathogen-free housing with a maximum of five mice per cage. The experiments were performed following the institutional guidelines strictly and were approved by the Institutional Animal Care and Use Committee of the Shanghai Model Organisms Center (2019-0011). CT26 cells ( $5 \times 10^5$ ) were intrasplenically injected at day 0, and a splenectomy was performed 30 minutes later. We then subcutaneously injected CT26 cells ( $3 \times 10^6$ ) at day 0, as previously described (6). Samples of primary tumors (subcutaneous) and metastatic tumors were collected at day 20. Four mice underwent liver metastasis and one mouse underwent colorectal peritoneal metastases. Those primary tumors ( $n = 4$ ) and metastatic tumors ( $n = 5$ ) were then sequenced by scRNA-seq.

## Single-Cell Isolation

We cut the fresh tissue into small pieces and placed them in RPMI 1640 medium (Gibco; 11875093) for 1 hour at 37°C. In detail, 1 mg/mL collagenase IV (Gibco; 17104019) and 0.4 mg/mL hyaluronidase were used for digestion. To dissociate the tumor tissues, we used the gentleMACS tumor dissociation kit (Miltenyi Biotec; 130095929). We then suspended and filtered cells through a 400- $\mu$ m mesh sieve. According to the manufacturer's instructions, flow cytometry was performed on a BD LSRFortessa cell analyzer (BD Biosciences) and analyzed using FlowJo software (V9.3.2), as we described previously (25).

## scRNA-seq

Following tissue digestion, samples were sequenced by 10x Chromium single-cell platform (10x Genomics, 30 v3 chemistry, Rev C). In detail, Gel Bead-In EMulsion (GEM) generation and barcoding was first performed. Then, post GEM-RT cleanup and cDNA amplification were conducted. Finally, the 3' gene-expression library construction was performed following the manufacturer's instructions. Completed libraries were sequenced on NovaSeq (Illumina). The Cell Ranger (V3) pipeline was used for demultiplexing, barcode processing, alignment, and initial clustering of the raw scRNA-seq profiles (25).

## ST

After resection, samples were collected, and a tissue optimization experiment was performed (10x Genomics, Visium Spatial Tissue Optimization, Rev A). In detail, tissue staining and imaging were first performed. Then, tissue permeabilization and fluorescent cDNA synthesis were performed, and RNA quality was checked (RIN > 7.0). Finally, tissue was removed, and slide imaging was performed. Thereafter, Visium Spatial Gene Expression was conducted (10x Genomics, Visium Spatial Gene-Expression Reagent Kits, Rev B). First, tissues were fixed and stained with imaging. Second, tissue permeabilization with different times (3, 6, 12, 18, 24, and 30 minutes) was performed, and tissue permeabilization with 12 minutes results in maximum fluorescence signal in both tumor regions and adjacent normal regions in the liver and colon. Next, reverse transcription, second-strand synthesis and denaturation, cDNA amplification and QC, Visium spatial gene-expression library construction, and ST sequencing were performed, respectively, following the manufacturer's instructions. This generated data sets containing 3,826 (ST-P1, liver), 4,658 (ST-P2, liver), 3,695 (ST-P3, liver), 3,721 (ST-P4, liver), 3,313 (ST-P1, colon), 4,174 (ST-P2, colon), 4,007 (ST-P3, colon), and 3,902 (ST-P4, colon) spots.

## mIHC

We performed the fluorescent dyes by using the Osteopontin/SPPI rabbit anti-human antibody (Abcam; catalog no. ab214050), Ki-67

rabbit anti-human antibody (IBP; catalog no. IR098), CD68 mouse anti-human antibody (Maxim; catalog no. Kit-0026), CCL18 rabbit anti-human antibody (Sino Biological; catalog no. 10502-T16, RRID:AB\_2860192), GranzymeK mouse anti-human antibody (ProteinTech; catalog no. 67272-1-Ig, RRID:AB\_2882541), GZMB rabbit anti-human antibody (Biolynx; catalog no. BX50024), CD56 rabbit anti-human antibody (IBP; catalog no. IR040), CTLA4 mouse anti-human antibody (Origene; catalog no. TA810299), CD45 mouse anti-human antibody (BioLegend; catalog no. 304002, RRID:AB\_2661811), CD15 mouse anti-human antibody (Maxim; catalog no. MAB-0779), DAPI (BioLegend; catalog no. 422801), MIF polyclonal rabbit anti-human antibody (Abcam, ab65869), GSTO1 rabbit anti-human antibody (Abcam, ab129106), IL4I1 rabbit anti-human antibody (Abcam, ab222102), SIRPA mouse anti-human antibody (OriGene, TA807751), CD47 rabbit anti-human antibody (Abcam, ab226837), and cytokeratin Pan (Pan Ck) mouse anti-human antibody (LBP Medicine, IHC-M067). Following the manufacturer's instructions (PerkinElmer, Opal Kit), we scanned the slides by using the PerkinElmer Vectra3 platform and quantified the results by using PerkinElmer Vectra3 platform as previously described (25, 78).

## scRNA-seq Data Processing, Cell-Type Annotation, Quality Control, and Data Visualization

Raw fastq files were first processed by Cell Ranger V3. Raw sequencing reads were then aligned to human genome (GRCh38, ENSEMBL). We then used Seurat (V3.2.2; refs. 22, 37) to process the unique molecular identifier (UMI) count matrix and integrates all cells according to sample ID by using Harmony (79). We performed the clustering analysis of all cells and only reserved the CD45<sup>+</sup> clusters. To define the main immune cell types, we then split the cells according to commonly used cell markers (CD4 for CD4 T cells, FOXP3 for Tregs, CD8A for CD8 T cells, SLC4A10 for MAIT cells, CD79A for B cells, MZB1 for plasma cells, LYZ for myeloid cells, and KLRC1 for NK cells; refs. 25, 80).

After labeling the main cell types, we performed clustering analysis of the split main cell populations. As for the CD4<sup>+</sup> cells, we found eight clusters. Cluster 1 was SELL<sup>+</sup> and was hence defined as CD4<sup>+</sup>-naïve T cells (81). CCL4, CCL5, and IFNG were highly expressed in cluster 2 (CCL5<sup>+</sup> CCL4<sup>+</sup> IFNG<sup>+</sup> CD4<sup>+</sup> T cells). Cluster 3 expressed CCL5 but not CCL4, so this cluster was classified as CCL5<sup>+</sup> CCL4<sup>-</sup> CD4<sup>+</sup> T cells. Given clusters 2 and 3 were similar, we directly combined them into a same cluster and labeled it as CCL5<sup>+</sup> CD4<sup>+</sup> T cells. Cluster 4 expressed TOX2, TIGIT, and PDCD1, and we defined it as PDCD1<sup>+</sup> CD4<sup>+</sup> T cells. Cluster 5 expressed KLRB1 and CCR6, and we defined it as CCR6<sup>+</sup> Th17 cells (82). Cluster 6 expressed LTB, and we defined it as LTB<sup>+</sup> CD4<sup>+</sup> T cells. Cluster 7 expressed HSP genes such as HSPA1A and HSP90AA1, and we defined it as HSP<sup>+</sup> CD4<sup>+</sup> T cells. Cluster 8 expressed FOXP3, TIGIT, and CTLA4, and we defined it as FOXP3<sup>+</sup> Tregs.

As for the CD8<sup>+</sup> cells, we found 10 clusters. Cluster 1 was GZMK<sup>+</sup>; hence, it was defined as GZMK<sup>+</sup> CD8<sup>+</sup> T cells. Cluster 2 expressed CD52; we hence defined it as CD52<sup>+</sup> CD8<sup>+</sup> T cells. Cluster 3 expressed SLC4A10, DPP4, and KLRB1 (83), and this cluster was defined as MAIT cells. Cluster 4 expressed CTLA4, TNFRSF9, HAVCR2, and TIGIT, so this cluster was classified as CTLA4<sup>+</sup> CD8<sup>+</sup> T cells (exhausted T cells; ref. 84). Cluster 5 expressed GIMAP family genes such as GIMAP7, GIMAP4, and GIMAP1. So we defined it as GIMAP7<sup>+</sup> CD8<sup>+</sup> T cells. Cluster 6 expressed SELL and CCR7, and we defined it as naïve CD8<sup>+</sup> T cells (81). Cluster 7 expressed FGFBP2 and GZMB, and we defined it as FGFBP2<sup>+</sup> GZMB<sup>+</sup> CD8<sup>+</sup> T cells (effector T cells; ref. 85). Cluster 9 expressed XCL1, and we defined it as XCL1<sup>+</sup> CD8<sup>+</sup> T cells. Cluster 10 expressed HSP genes such as HSPA6, HSPA1A, HSPD1, and HSPH1, and we defined it as HSP<sup>+</sup> CD8<sup>+</sup> T cells. In addition, clustering analysis also generates a cluster with high expression of TRDC but low expression of the CD8A and CD4 genes. We hence define it as TRDC<sup>+</sup>  $\gamma\delta$  T cells (86).

As for the NK cells, we found four clusters. Cluster 1 expressed *GZMK* and *XCL1* and was hence named as *GZMK*<sup>+</sup> NK cells. Cluster 2 expressed *GZMK*, *IL7R*, and *GPR183*, and was hence defined as *GZMK*<sup>+</sup> *IL7R*<sup>+</sup> *GPR183*<sup>+</sup> NK cells. Cluster 3 expressed *GZMB* and *GNLY* and was hence named as *GZMB*<sup>+</sup> NK cells. Cluster 4 expressed *GZMB* and *MYOM2*. Thus, it was defined as *GZMB*<sup>+</sup> *MYOM2*<sup>+</sup> NK cells. Given that clusters 1 and 2 both expressed *GZMK*, we hence combined them into the same cell cluster and labeled it as *GZMK*<sup>+</sup> NK cells. Given that clusters 3 and 4 both expressed *GZMB*, we combined them into the same cell cluster and labeled it as *GZMB*<sup>+</sup> NK cells.

As for the myeloid cells, there existed nine clusters. Cluster 1 was *MKI67*<sup>+</sup> and was hence defined as *MKI67*<sup>+</sup> proliferation macrophages. *S100A8*, *S100A9*, and *S100A12* were highly expressed in cluster 2 (*S100A8*<sup>+</sup> monocytes; ref. 8). Cluster 3 expressed *CLEC10A*, and this cluster was defined as *cDC2* cells (87). Cluster 4 expressed *FCGR3B*, so this cluster was classified as neutrophils (88). Cluster 5 expressed HSP genes such as *HSPA1B*, *HSPH1*, *HSPA1A*, and *HSPB1*, and we defined it as HSP<sup>+</sup> monocytes. Cluster 6 expressed *MRC1* and *CCL18*, and we defined it as *MRC1*<sup>+</sup> *CCL18*<sup>+</sup> M2-like macrophages (89). Cluster 7 expressed *CLEC9A*, and we defined it as *CLEC9A*<sup>+</sup> *cDC1* (8). Cluster 8 expressed *SPPI* and we defined it as *SPPI*<sup>+</sup> macrophages (8). Cluster 9 expressed *CXCL10*, and we defined it as *CXCL10*<sup>+</sup> M1-like macrophages (89).

As for the B cells, cluster 1 was *MKI67*<sup>+</sup> and was hence defined as *MKI67*<sup>+</sup> proliferation B cells. Cluster 2 expressed *AIM2*, and this cluster was defined as *AIM2*<sup>+</sup> memory B cells. Cluster 3 expressed *TCL1A* and *SELL*, so this cluster was classified as *TCL1A*<sup>+</sup>-naïve B cells (90). Cluster 4 expressed *EGRI*, which is essential for B-cell activation (91), and was labeled as *EGRI*<sup>+</sup>-activated B cells. Cluster 5 expressed HSP genes such as *HSPA1A*, *HSP90AA1*, *HSPE1*, and *HSPH1*, and we defined it as *HSPA1A*<sup>+</sup> B cells. Cluster 6 expressed a broad range of IGHA genes such as *IGHA2* and *IGHA1*. We hence labeled it as *IGHA*<sup>+</sup> plasma B cells. Cluster 6 showed high expression of *IGHG1*, *IGHG2*, *IGHG4*, and *IGHG3*. This indicates that such cluster is *IGHG*<sup>+</sup> plasma B cells.

As for quality control of scRNA-seq data, we set the cutoff min. cells as 3 and min.features as 500 in the CreateSeuratObject function in Seurat (V3.2.2). To remove the potential doublets, we perform two rounds of doublet removal by using doubletFinder (V2.0.3; ref. 92). We first run doubletFinder on all cells and then perform similar analysis on subcellular populations. When performing clustering analysis, we remove the cluster with high expression of mitochondrial genes (93).

To visualize our data, we use Seurat (V3.2.2) (22), ggplot2 (V3.3.2), dittoSeq (V1.1.10) (94), ggrepel (V0.8.2), ggpubr (V0.4.0), ggsignif (V0.6.0), pheatmap (V1.0.12), circlize (V0.4.10), and cowplot (V1.1.0).

### Cellular Proportion Analysis

To perform the differential cell-type composition analysis, we utilized the scDC algorithm (23) to conduct the bias-corrected and accelerated bootstrap analysis for cell-type proportion comparison. We used the scDC\_noClustering function to perform the bootstrap resampling cell-type composition analysis (calCI = TRUE, calCI\_method = BCa, nboot = 1,000). To find out the tissue-specific cellular types, we used the median value of bootstrap resampled cellular composition, filtered the immune cells with low composition, and performed the unsupervised clustering analysis.

### ST Data Processing and Cell-Type Estimation

We used Space Ranger V1.2.1 to process raw fastq files. We next used Seurat (V3.2.2; ref. 22) to process the Space Ranger output files. In detail, we used SCTransform (95) to normalize data, RunPCA to perform dimension reduction, FindNeighbors and FindClusters to cluster the ST spots, and RunUMAP to visualize the data. Given that each spot contains several cells (74), we next

combined the scRNA-seq data and predicted main cell types by using Seurat (95). As for subpopulation annotation, we found that Seurat did not generate uniform results, which might be due to the overlapping cell markers among subcellular populations. We hence used ssGSEA (96) to score the subcell types, which has been proved to be robust for ST data (29).

### Bulk RNA-seq, Microarray Analysis, and Survival Analysis

We fetched bulk RNA-seq data of patients with colorectal cancer from the TCGA (<https://portal.gdc.cancer.gov/>). We next used ssGSEA (96) to estimate the immune cell infiltration level by using specific cell markers (avg\_logFC > 0.25, pct.1 > 0.5, and pct.2 < 0.5). We next used survival (V3.2-7) and survminer (V0.4.8) to generate the Kaplan–Meier plot, and the *P* value was calculated through the log-rank test. To validate the cell infiltration level, we used another CRLM microarray data set (ref. 30; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE41568>) and utilized ssGSEA to estimate the immune cell level.

### Cell Differentiation Trajectory Inference

To infer the differentiation trajectory of intratumor myeloid cells, we first used monocle (V2.14.0; ref. 97) to infer the pseudotime of each cell (method = “DDRTress,” ordering\_genes = marker genes). To compare the myeloid cell lineage/clustering difference between untreated samples and NAC-treated samples, we used dyno (V0.1.2, method = “slingshot”; refs. 57, 98) to analyze and visualize.

### Intercellular Ligand–Receptor Analysis

We used CellPhoneDB V2.0 to infer the ligand–receptor cross-talk between single cells (99). As for the differential tumor–immune cell cross-talk analysis (colorectal cancer vs. liver metastasis), we separately computed the tumor–immune cell cross-talk in colorectal cancer and liver metastasis. We selected the significantly differential cross-talk ligand–receptor pairs and used circlize (V0.4.10) to visualize the results.

### Differential Gene Expression and Pathway Enrichment Analysis

We used FindMarkers function to perform differential gene-expression analysis of different cellular populations and set *P*<sub>adj</sub> = 0.05 as the cut-off value. We next performed the pathway enrichment analysis of those differentially expressed genes by using clusterProfiler V3.19.0 (refs. 43, 100; enricher function, KEGG gene sets, or cancer hallmark gene sets from msigdb). We set *P* = 0.05 as the cut-off value.

### Clustering Single Cells Using TooManyCells

To compare the cellular clustering difference between untreated samples and NAC-treated samples, we separately performed clustering analysis on colorectal cancer and liver metastasis samples by using TooManyCells (55). In detail, we used too-many-cells make-tree function and set draw-collection as “PieRing.”

### Development of scMetabolism R Package

scMetabolism was designed to easily quantify single-cell metabolic activity by using single-line command. Before metabolism gene signature scoring, users can choose whether to impute their scRNA-seq matrix (38). The core function of scMetabolism is to quantify metabolic pathway gene sets. We combined the published gene sets (36) and manually reviewed gene sets from the KEGG database (42, 43) and the REACTOME database (43, 44) to generate the list of metabolic gene sets. As for the quantification method, scMetabolism currently supports VISION (39), AUCell (40), and ssGSEA (41). Users

can directly use the scMetabolism pipeline based on their Seurat object (22, 37).

### scMetabolism Performance Evaluation

To evaluate the performance of scMetabolism, we used four different data sets, including PBMCs, melanoma, head and neck squamous cell carcinoma, and testis (45–47). To test the accuracy of this pipeline, we compared VISION and AUCell with ssGSEA, which was previously reported to be appropriate for single-cell metabolism quantification (36). We found that VISION shows good correlation with ssGSEA, with the median Spearman  $\rho$  estimate over 0.8 in all data sets (Supplementary Fig. S4A). As for AUCell, its correlation with ssGSEA ranges from 0.5 to 0.8 (Supplementary Fig. S4A). Hence, ssGSEA and VISION show best consistency among all methods.

Recent pipeline from Locasale Lab by default imputes the missing gene-expression values. However, it still remains unclear how dropout affects the metabolic quantification and whether imputation can improve the performance. We hence simulated the dropout events by randomly select 5%, 10%, 20%, and 30% expressed genes and set their values to 0. In those four simulated dropout gene-expression matrix, we used ALRA (38) to impute the simulated missing data. Next, we used VISION, AUCell, and ssGSEA to quantify the metabolic activity and perform the correlation analysis with the raw data matrix. Interestingly, we found imputation may not improve the metabolic quantification performance (Supplementary Fig. S4B–S4E). In the 5% dropout-imputation data set, we observed a sharp decreased correlation value, whereas the correlation remains similar in the 30% dropout-imputation data set in all data sets. Among all three methods, results of VISION and ssGSEA are less affected by dropout. Among all methods, VISION is the fastest way for metabolism quantification. We hence set VISION as the default method and set “not imputation” as the default option. Finally, we developed a web server, scMetabolism online (<http://www.cancerdiversity.asia/scMetabolism/>), for experimental biologist to easily perform metabolism quantification.

### Web Server Development

We used shiny (V1.15.0) to construct the web server and runs on a Linux-based Apache Web server (<http://www.apache.org>) as previously described (101, 102). We tested this web server on different web browsers, including Chroma, Safari, and Internet Explorer. Users are not required to register or login to access features in the database.

### Data and Code Availability

This raw sequencing data generated during this study are available at The National Omics Data Encyclopedia (<https://www.biosino.org/node/project/detail/OEP001756>). The analyzed data are publicly available in the web portal (<http://www.cancerdiversity.asia/scCRLM>). The R package of scMetabolism is available on Github (<https://www.github.com/wu-yc/scMetabolism>) and can be directly installed. The online version of this package, named as scMetabolism online, is also publicly available (<http://www.cancerdiversity.asia/scMetabolism>).

### Authors' Disclosures

No disclosures were reported.

### Authors' Contributions

**Y. Wu:** Software, formal analysis, validation, investigation, visualization, methodology, writing—original draft, writing—review and editing. **S. Yang:** Software, formal analysis, validation, investigation, visualization, methodology, writing—original draft, writing—review and editing. **J. Ma:** Software, formal analysis, validation,

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